Preserving Hamiltonian Symmetry on Parallel Hardware: Deterministic Reductions and Mixed-Precision Acceleration in GPU Symplectic Integration

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***Abstract*—Symplectic integrators keep the discretized flow on the same side of Noether's theorem as the continuous system: the conserved energy oscillates within a bounded band instead of drifting. GPUs promise throughput for such kernels but threaten both halves of the promise. Reduction order on the device is not fixed, so the reported conserved quantity varies bit-for-bit between runs, and FP32—64× cheaper than FP64 on consumer silicon—quantizes the very oscillation the method preserves. We present *noether*, a CUDA engine for conservative orbital dynamics (Kepler plus zonal harmonics J_2–J_4) whose Verlet, Yoshida-4, and RK4 kernels share one templated force evaluation, with three interchangeable energy reductions (atomic, fixed merge-tree, Kahan) crossed against FP64/FP32 accumulators. On an RTX 2050, fixed and Kahan reductions are bitwise reproducible over 100 repeats for every $n \le 2^{24}$, while atomic summation yields up to 42 distinct FP64 results; an adversarial input drives FP32 sums to exactly zero (100% error) where FP64 stays exact; and the FP32 accumulator noise floor, 6.5×10^{-8} in relative energy, sits above the 5×10^{-8} symplectic oscillation it would have to resolve. A mixed-precision Verlet—FP32 state, persistent Kahan compensation on the kick, FP64 energy evaluation—delivers 6.7× GPU throughput (13× over a 12-thread CPU) with a 7-day relative energy error of 6.5×10^{-6}.**

***Index Terms*—Hamiltonian mechanics, symplectic integration, GPU computing, floating-point arithmetic, reproducibility.**

I. Introduction

An autonomous mechanical system with Hamiltonian H conserves energy by Noether's theorem; a *symplectic* numerical integrator preserves the corresponding structure of the map, and the practical signature of that preservation is a bounded, oscillatory error $\Delta H(t)$ rather than secular drift [3]. This makes energy error both the quality metric and the diagnostic: it is cheap to evaluate and it separates good discretizations from merely stable ones.

Parallel hardware complicates the picture in two ways. First, the final summation that turns per-particle energies into the reported H is itself a floating-point reduction, and on a GPU its order is not fixed: the same input can produce a different last bit on two consecutive runs, so a quantity that is supposed to be a diagnostic becomes a random variable. Second, consumer GPUs price precision asymmetrically—our RTX 2050 executes one FP64 FMA per 64 FP32 lanes—so the natural way to buy throughput is to shrink state or accumulators to FP32, which quantizes H at exactly the scale of the oscillation one is trying to observe.

This paper presents *noether*, a compact C++20/CUDA engine for conservative orbital dynamics (Kepler gravity plus the zonal harmonics J_2–J_4) and the measurement campaign behind three claims:

1) **Determinism is a reduction-mode choice.** A fixed merge-tree or Kahan reduction is bitwise reproducible at every size up to $n = 2^{24}$; atomic summation is not, even under one fixed launch configuration, and the spread varies with n (Sec. V-B).
2) **FP32 reductions have a hard noise floor.** With FP32 accumulators, the relative-energy ULP is 6.5×10^{-8}, larger than the 5×10^{-8} symplectic oscillation itself, and adversarial cancellation can erase the sum entirely (Secs. V-B–V-C).
3) **Mixed precision is a Pareto trade, not a compromise.** An FP32 Verlet with persistent Kahan compensation on the velocity kick runs 6.7× faster than FP64 on the GPU and 13× faster than the multithreaded CPU while keeping 7-day $\Delta H/H_0$ at 6.5×10^{-6} (Sec. V-D).

II. Background

A. Symplectic integration of orbital motion

We integrate states $[x, y, z, v_x, v_y, v_z]$ under an autonomous Hamiltonian $H = \sum_i \frac{1}{2}\|v_i\|^2 + \Phi(r_i)$. Velocity-Verlet (kick–drift–kick) is second order and symplectic for separable Hamiltonians; Yoshida's composition [2] lifts it to fourth order with three substeps (w_0, w_1, w_1) at times $(0, c_2, 0)$ where $c_2 = \sqrt[3]{2}$, $w_1 = (2 - c_2)^{-1}$, $w_0 = -c_2/(2 - c_2)$. Classical RK4 [3] serves as the non-symplectic control. Measured error reduction when halving Δt is 16.59× (RK4), 4.00× (Verlet), 15.98× (Yoshida-4), against theoretical 16/4/16 (`tests/test_integrators.cpp`).

B. Order-dependent summation

Rounding error in a floating-point sum depends on evaluation order; Kahan's compensated summation [4] trades a serial dependency for $O(u^2)$ rather than $O(nu)$ error [5]. Deterministic parallel reductions fix the order by construction (merge trees, fixed chunking), at the cost of extra passes; atomics fix neither order nor reproducibility. Reproducibility is now a recognized axis of benchmark design [6]. Our contribution is to measure all three modes inside a physics kernel where the reduced value *is* the physical diagnostic, rather than a downstream linear-algebra result.

III. Engine Design

A. Conservative force model

The geopotential is evaluated in closed form, $\Phi(r) = -\frac{\mu}{r}\left[1 - \sum_{n=2}^{4} J_n \left(\frac{R_E}{r}\right)^n P_n(z/r)\right]$, and the acceleration is $-\nabla\Phi$, enforced component-wise against central finite differences at relative tolerance 10^{-6} for each harmonic (`tests/test_force.cpp`). Auditing against the gradient exposed and fixed a sign error in the legacy implementation's J_3 term (legacy J_3 was $-1\times$ the standard EGM96 value); two-body energy tests had been sign-insensitive to it. Drag, solar radiation pressure, and third-body terms are deliberately excluded: they make H non-conservative and destroy the measured quantity.

B. Scalar-templated integrators

Force and all three integrators are templates on the state scalar type (`double` or `float`), with the operation order written to be identical for both instantiations. The `double` instantiation is regression-checked bit-for-bit against pre-refactor reference trajectories, so switching precision changes the arithmetic type only, never the algorithm. On the device, states live in structure-of-arrays; stepping is fixed-step (warp uniformity forbids per-thread adaptive steps).

C. Reductions: atomic, fixed tree, Kahan

Every device reduction writes per-block partials through one templated kernel `k_partial<Acc,Kahan>` (Acc $\in$ {double, float}, 256 threads/block) and differs only in finalization: **atomic** adds partials to a shared accumulator with `atomicAdd` (one pass, order varies run to run); **fixed** merges partials through an out-of-place halving tree with a fixed slot layout (order fully determined by n and block count); **Kahan** carries a compensation value through both the local partial and the fixed merge. The same trio is used for per-particle Hamiltonians and for log-style scalar reductions; the CLI exposes mode, accumulator precision, and stream count as independent knobs.

D. Mixed-precision Verlet

The FP32 path uses a separate ensemble type holding `float` states plus three persistent compensation arrays (c_x, c_y, c_z per particle) that are zeroed once at upload, not per step. The velocity kick accumulates in FP32 with Kahan compensation—the kick is where rounding re-enters every step—while drift and state storage stay plain FP32; energies are always evaluated in FP64 from the FP32 state. Because the compensation persists across steps, rounding error is carried rather than re-rounded, which is what separates the `fp32k` variant from a per-step compensated sum.

IV. Experimental Setup

All measurements run on a laptop RTX 2050 (SM86, FP64 rate 1/64 FP32) with CUDA 12.9, compiled by g++ 14.2 ($-O3$, C++20, no fast-math flags, so FP64 paths remain IEEE-conformant); the CPU reference is the same binary with OpenMP across 12 threads. Bodies orbit a circular 400 km, 51.6°-inclination orbit; ensembles spread RAAN and argument of latitude over a deterministic golden-ratio sequence (`core/ensemble.hpp`). The CPU reference accumulates H over bodies in long double; GPU reductions accumulate in FP64/FP32 partials. Four experiments produce every number and figure below, all regenerable by `scripts/run_experiments.sh` into `paper/results/`:

- **E1 Integrator structure:** RK4/Verlet/Yoshida-4, $\Delta t \in \{10, 20, 40\}$ s, 24 h and 7 d, CPU, sampled every 600 s (18 CSVs, `e1_*`).
- **E2 Reduction repeatability:** $n = 2^{10} \ldots 2^{24}\times$ benign LCG and adversarial inputs $\times$ 6 mode/precision combinations $\times$ 100 repeats, single stream (9 600 rows, `e2_reduce.csv`).
- **E3 Noise floor:** 7 d ensemble drift, $n = 4096$, $\Delta t = 10$ s, every reduction mode on GPU vs. CPU reference, sampled every 300 s (7 CSVs, `e3_*`).
- **E4 Precision Pareto:** FP64/FP32/FP32k Verlet on GPU, $\Delta t \in \{5, 10, 20, 40\}$ s, 7 d (12 CSVs, `e4_*`) plus throughput, $n = 2^{20}$, 200 steps, best of 5.

V. Results

A. Structure versus time step

Figure 1 shows the 7-day energy history. At $\Delta t = 10$ s the symplectic methods hold a flat band while RK4 grows: maximum $\Delta H/H_0$ is 3.56×10^{-9} (RK4), 1.04×10^{-7} (Verlet), 2.88×10^{-11} (Yoshida-4), and RK4's daily envelope expands from 5.0×10^{-10} on day 1 to 3.6×10^{-9} on day 7. The methods cross over as Δt grows: at $\Delta t = 40$ s, RK4's 3.64×10^{-6} *exceeds* Verlet's 1.18×10^{-6}, while Yoshida-4 stays two orders below both (7.38×10^{-9}). The classical picture—error $\propto \Delta t^p$—holds in the small-step regime; the practically relevant ranking at coarse steps is bounded-versus-growing, not order.

B. Reduction determinism and accuracy

Figure 2 summarizes all 9 600 repeats. On the benign input, FP64 accumulators sit at the summation floor (1.2×10^{-17} to 3.6×10^{-15} relative over all modes and sizes); FP32 errors span 1.8×10^{-9} to 7.4×10^{-8} for the fixed and Kahan trees and reach 3.0×10^{-6} for atomic summation. The adversarial input ($\pm10^8$ with an i mod 7 residue) is an exact cliff: every FP32 summation—all three modes, all sizes, all 100 repeats—rounds to exactly zero (relative error 1.0), whereas FP64 partial

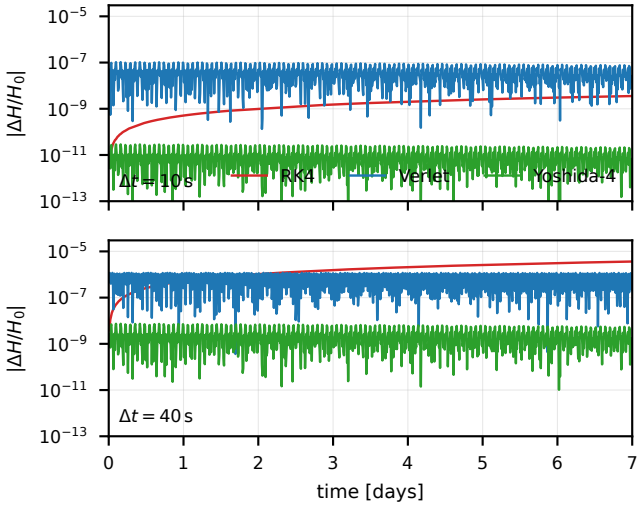


Fig. 1. 7-day relative energy error, CPU, $\Delta t = 10$ s (top) and 40 s (bottom). RK4 (dashed) grows secularly; Verlet and Yoshida-4 remain bounded. Source: `paper/results/e1_*_7d.csv`.

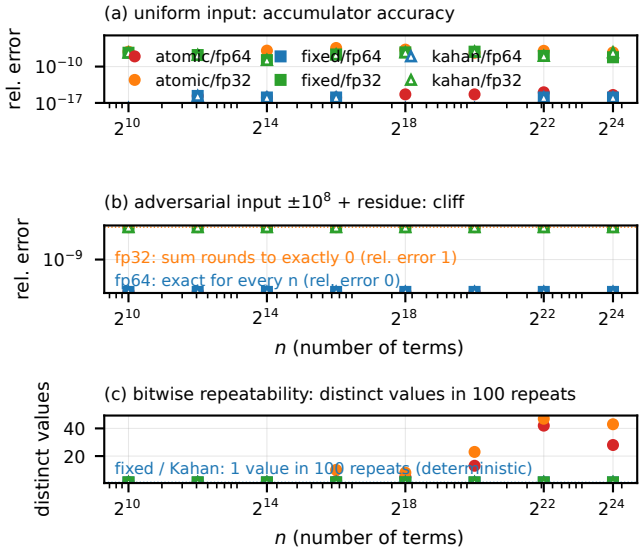


Fig. 2. Reduction quality over 100 repeats. (a) Benign input accuracy ladder; (b) adversarial cliff—FP32 collapses to a zero sum, FP64 stays exact; (c) distinct results in 100 repeats—fixed/Kahan always one, atomic varies with n. Source: `paper/results/e2_reduce.csv`.

sums stay integral below 2^{53} and are bit-exact against the long-double oracle.

Repeatability is a mode property, not a platform accident (Fig. 2c). Fixed-tree and Kahan reductions return exactly one distinct value in 100 repeats at every n and both accumulator precisions. Atomic summation returns 2, 5, 13, and 42 distinct FP64 values at $n = 2^{16}, 2^{18}, 2^{20}, 2^{22}$ (28 at 2^{24}), and 8–47 distinct FP32 values at $n \geq 2^{16}$—under one fixed launch configuration, so the variation comes from device-side scheduling of atomic adds, not from our configuration knobs. The cost of determinism is modest at $n = 2^{20}$: 0.342 ms fixed and 0.405 ms Kahan versus 0.309 ms atomic for an FP64 reduction (11% and 31% overhead).

C. The FP32 noise floor

Figure 3 overlays the first day of the 7-day ensemble experiment. All three FP64 reduction modes and the long-double CPU reference are indistinguishable: identical range $[-5.18, +2.66] \times 10^{-8}$, 100% of samples nonzero. With FP32 accumulators the physics disappears into arithmetic. Summing $H \approx -1.2 \times 10^5$ in FP32 has ULP $= 2^{-7} = 7.8 \times 10^{-3}$, i.e. a relative quantum of 6.49×10^{-8}—larger than the oscillation itself. Fixed and Kahan FP32 tick at exactly that quantum (range $\pm 6.49 \times 10^{-8}$, 69% of samples nonzero), and atomic FP32 pins a single value for the entire run (every sample identically zero relative energy error). This is not a bug in the atomic path: no FP32 accumulator, in any mode, can represent the quantity being measured.

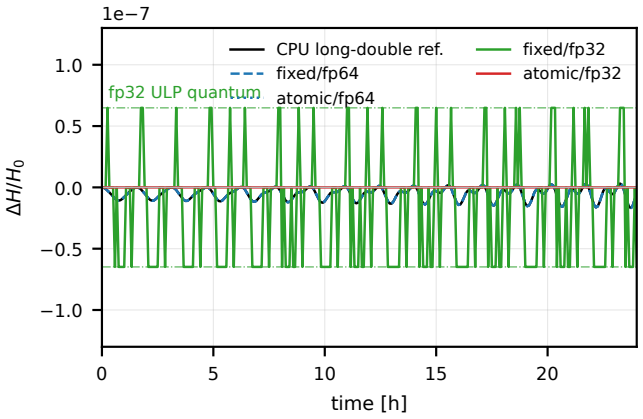


Fig. 3. First 24 h of E3 ($n = 4096$): FP64 modes track the CPU reference; FP32 modes are quantized at the $\pm 6.49 \times 10^{-8}$ ULP quantum (dash-dot), with atomic FP32 flat at zero; Kahan FP32 is bit-identical to fixed FP32 (not drawn). Source: `paper/results/e3_*.csv`.

D. Accuracy-throughput Pareto

Table I gives 7-day accuracy for the three state-precision variants; Fig. 4 places them against measured throughput. FP64 GPU Verlet runs at 4.82 ms/step versus 9.60 ms for the 12-thread CPU (1.99×). FP32 state runs at 0.483 ms (9.98× over GPU FP64, 19.9× over CPU) and the compensated variant at 0.724 ms (6.65×, 13.3×). The compensation earns its cost at every time step: FP32k beats FP32 everywhere, by 4.2× at $\Delta t = 10$ s (6.48×10^{-6} vs. 2.72×10^{-5}). Note the honest inversion of the classical rule for FP32: error *falls* as Δt grows (2.8×10^{-5} at 5 s to 6.7×10^{-6} at 40 s) because per-step rounding, not truncation, dominates at small Δt. Figure 4 shows the resulting bands: precision sets the vertical position, state precision sets the horizontal one, and FP32k occupies the useful corner.

TABLE I
MAX $|\Delta H/H_0|$ OVER 7 D, GPU VERLET (E4).

Δt (s)	5	10	20	40
FP64	2.69×10^{-8}	1.04×10^{-7}	3.68×10^{-7}	1.18×10^{-6}
FP32	2.80×10^{-5}	2.72×10^{-5}	2.09×10^{-5}	6.72×10^{-6}
FP32k	1.56×10^{-5}	6.48×10^{-6}	1.64×10^{-5}	4.40×10^{-6}

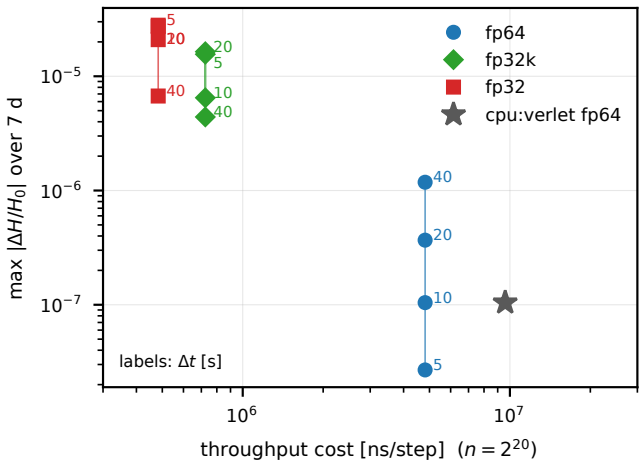


Fig. 4. 7-day max $|\Delta H/H_0|$ versus time per step ($n = 2^{20}$, best of 5). Each precision forms a vertical band over $\Delta t \in \{5, 10, 20, 40\}$ s; the star marks the 12-thread CPU. Source: `paper/results/e4_*.csv`.

VI. Threats to Validity

Hardware generality. All throughput numbers come from one laptop Ampere GPU whose FP64 rate is 1/64 FP32; on datacenter parts (e.g. 1:2 rates) the FP32 speedups shrink substantially, while the determinism and noise-floor results—properties of IEEE arithmetic and reduction order—do not. **Model scope.** The engine models a conservative field only; dissipative forces would change H by design and make this diagnostic meaningless, so conclusions apply to energy-conserving settings. **Coverage.** One orbit regime (400 km, 51.6°), $n \le 2^{24}$, fixed-step integrators, and one device; atomic spread is measured under a single launch configuration, and other GPUs or chunking schemes will show different—but almost certainly nonzero—spread. **Build.** No fast-math flags were used; results under flush-to-zero modes or different compilers may shift FP32 floors by an ULP or two, not their order of magnitude.

VII. Conclusion

Noether's correspondence—symmetry, conservation, diagnostic—survives parallel hardware only if the reduction that reports the conserved quantity is treated as part of the algorithm, not as plumbing. Fixed-tree and Kahan reductions restore bitwise determinism at 11–31% overhead; atomic summation does not, and its run-to-run spread varies with problem size. FP32 accumulation is not a cheaper version of FP64: it changes what can be measured at all, quantizing the symplectic oscillation into ULP steps and collapsing under cancellation. The useful operating point on this hardware is mixed precision: FP32 state with persistent kick compensation and FP64 energy evaluation buys 6.7× GPU throughput at a bounded 6.5×10^{-6} error, leaving the structure of the flow—and the diagnostic—intact.

Artifact Appendix

Everything is reproducible from a clean clone with three commands:

1) `cmake -S . -B build -DCMAKE_BUILD_TYPE=Release && cmake --build build -j && ctest --test-dir build --output-on-failure` (5/5 tests; CUDA tests skip without a device).
2) `./scripts/run_experiments.sh` regenerates all 39 CSVs in `paper/results/` (~90 s with a GPU).
3) `python3 analysis/fig1_drift.py` (and `fig2`, `fig3`, `fig4`) regenerates `paper/figs/*.pdf`; requirements in `analysis/requirements.txt`.

Every number in this paper is computed from those CSVs by the figure scripts or the commands shown in the captions. The paper itself builds with `cd paper && tectonic main.tex` (any TEX toolchain with the `newtx` fonts works).

References

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